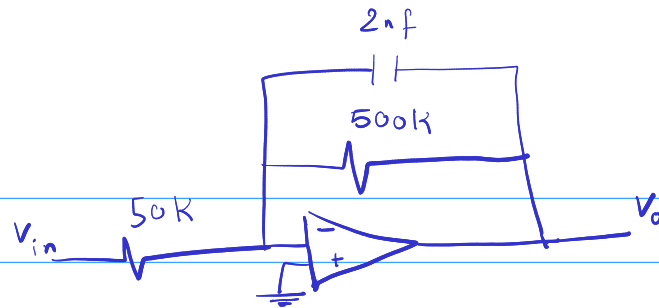
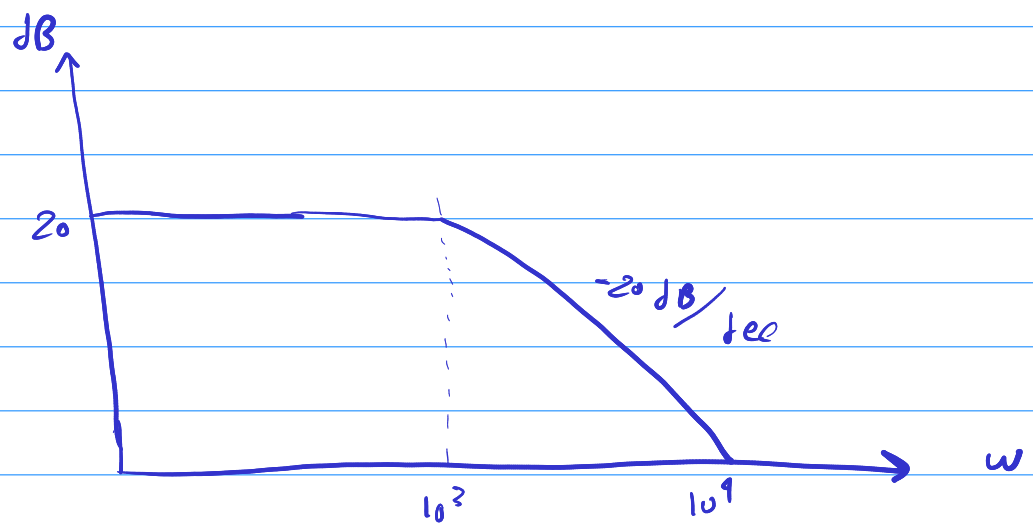


Q (2)



$$H(s) = \frac{-(500k // x_c)}{50} \rightarrow \frac{-\left(\frac{500k/sC}{500k + \frac{1}{sC}}\right)}{50k} = \frac{-10}{\left(\frac{s}{10^7} + 1\right)}$$

$$g = 20 \log(10) = 20$$

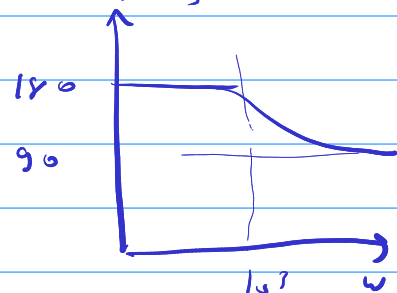


(b) $H(s) = 1 = \frac{\sqrt{(-10)^2 + 0}}{\sqrt{1^2 + \left(\frac{\omega}{10^3}\right)^2}} \rightarrow \omega = 9.95 \times 10^7 \text{ rad/s}$

(c) $\angle H(s) = \tan^{-1}\left(\frac{0}{-10}\right) - \tan^{-1}\left(\frac{\omega}{10^2}\right)$ degrees

@ $\omega = 0 \rightarrow \angle H(s) = 180 - 0 = 180^\circ$

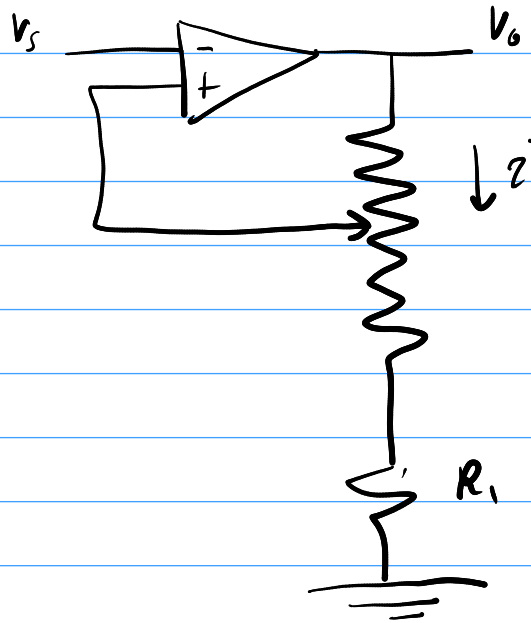
@ $\omega = \infty \rightarrow \angle H(s) = 180 - 90 = 90^\circ$



(d) $|T_1| = \frac{10}{\sqrt{1 + \left(\frac{\omega}{10^3}\right)^2}}, |T_2| = \frac{10}{\sqrt{1 + \left(\frac{\omega}{10^4}\right)^2}}$

$$20 \log(T_2) - 20 \log(T_1) \rightarrow \Delta T_{dB} = -17 \text{ dB}$$

Q (4)



$$i' = \frac{V_o}{R_1 + 100k} = \frac{V_s}{R_1 + 100k} \rightarrow T = \frac{R_1 + 100}{R_1 + 100k}$$

a) $R_1 = 0 \rightarrow T = \frac{1}{1000}$

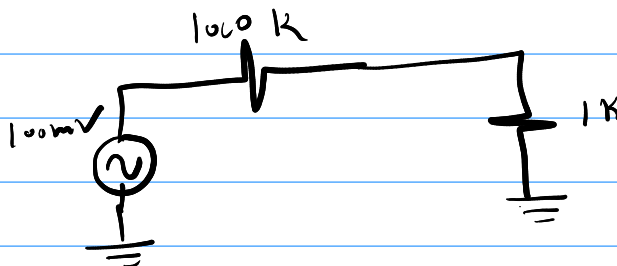
b) @ max $\rightarrow T = \frac{R_1 + 100k}{R_1 + 100k} = 1$

@ min $\rightarrow T = \frac{R_1 + 100k}{R_1} = 10$

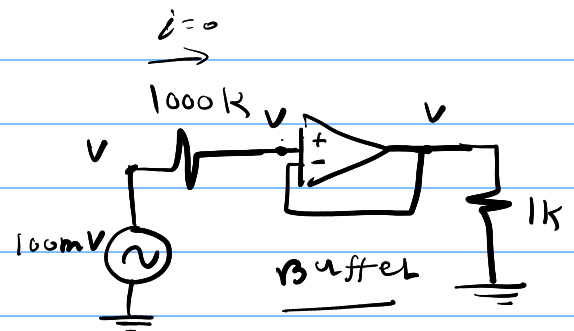
$$R_1 = 11.1k\Omega$$

Q (5)

a)



b)



0% drop

$$V_1 = \frac{1}{100+1} (100) \times 10^{-3}$$

$$V_1 = 0.99mV$$

99% drop